Performance	Below 20	21-30	Above 30	Total
Average	138	83	64	285
Good	64	67	84	215
Total	202	150	148	500

Solution:

Step 1 : **Null hypothesis** H_0: The attributes age and performance are independent.

Alternative hypothesis H_1: The attributes age and performance are not independent.

Step 2 : **Data**

Compute expected frequencies E_{ij} corresponding to each cell of the contingency table, using the formula

$$E_{ij} = \frac{R_i \times C_j}{N} \quad i = 1, 2; j = 1, 2, 3$$

where,

N = Total sample size

R_i = Row sum corresponding to i^{th} row

C_j = Column sum corresponding to j^{th} column

Performance	Below average	Average	Above average	Total
Average	$\frac{285\times202}{500}=115.14$	$\frac{285\times150}{500}=85.5$	$\frac{285\times148}{500}=84.36$	285
Good	$\frac{215\times202}{500}=86.86$	$\frac{215\times150}{500}=64.5$	$\frac{215\times148}{500}=63.64$	215
Total	202	150	148	500

Step 3 : **Level of significance** α = 5%

Step 4 : **Calculation**

Calculate the value of the test statistic as

$$\chi_0^2 = \sum_{i=1}^{2}\sum_{j=1}^{3}\frac{(O_{ij}-E_{ij})^2}{E_{ij}}$$

This chi-square test statistic is calculated as follows:

$$\chi_0^2 = \frac{(138-115.14)^2}{115.14} + \frac{(83-85.50)^2}{88.50} + \frac{(64-84.36)^2}{84.36} + \frac{(64-86.86)^2}{86.86} + \frac{(67-64.50)^2}{64.50} + \frac{(84-63.64)^2}{63.64}$$

= 22.152 with degrees of freedom (3–1)(2–1) = 2

Step 5 : **Critical value**

From the chi-square table the critical value at 5% level of significance is $\chi^2_{(2-1)(3-1),0.05} = \chi^2_{2,0.05} = 5.991$.

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